Application Note

Shear Wave Visualization Script
Description and Operation

Verasonics Inc. | www.verasonics.com | support@verasonics.com
11335 NE 122nd Way, Suite 100, WA 98034 USA

Important Notes

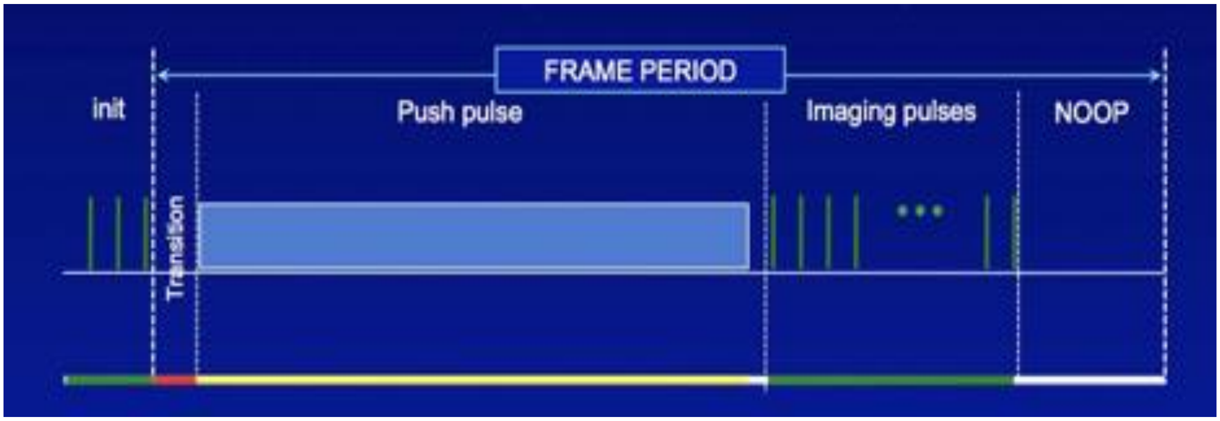
Before starting, please consider the following important notes.

1. This script is only used to visualize shear wave propagation, but no quantitative measurement, such as velocity or elasticity, is provided here.

2. ExtendBL (Profile 5) option: The ExtendBL option is required for the long pulse. The use of different power supply settings and/or higher transmit voltages could result in permanent damage to the transducer. If a user damages their system and it is determined that unauthorized modifications to the limit algorithms had been incorporated, Verasonics will not be responsible for the cost of any repairs even if the system is still under warranty or a service contract.

1. Introduction

The example script, SetUpL7_4ShearWaveImaging.m, is used to visualize shear wave propagation in realtime. Two sets of events are predefined and the VSX will execute either one based on the startevent setting in the event list. The first set is regular flash imaging used to determine the target for pushing. The second set is used to generate shear wave and store the IQData of each detection pulse following the sequence order below:



2. How to use it?

Some important parameters should be defined first:

na: number of detection acquisitions, i.e., the number of pages in an InterBuffer frame.

SWIFrames: number of detection frames. Each SWIFrame has na acquisitions.

BmodeFrames: number of regular flash image frames.

Since the filename and VSX have been included in the script, VSX will be executed automatically after running the script and the default GUI will appear with regular flash imaging mode:

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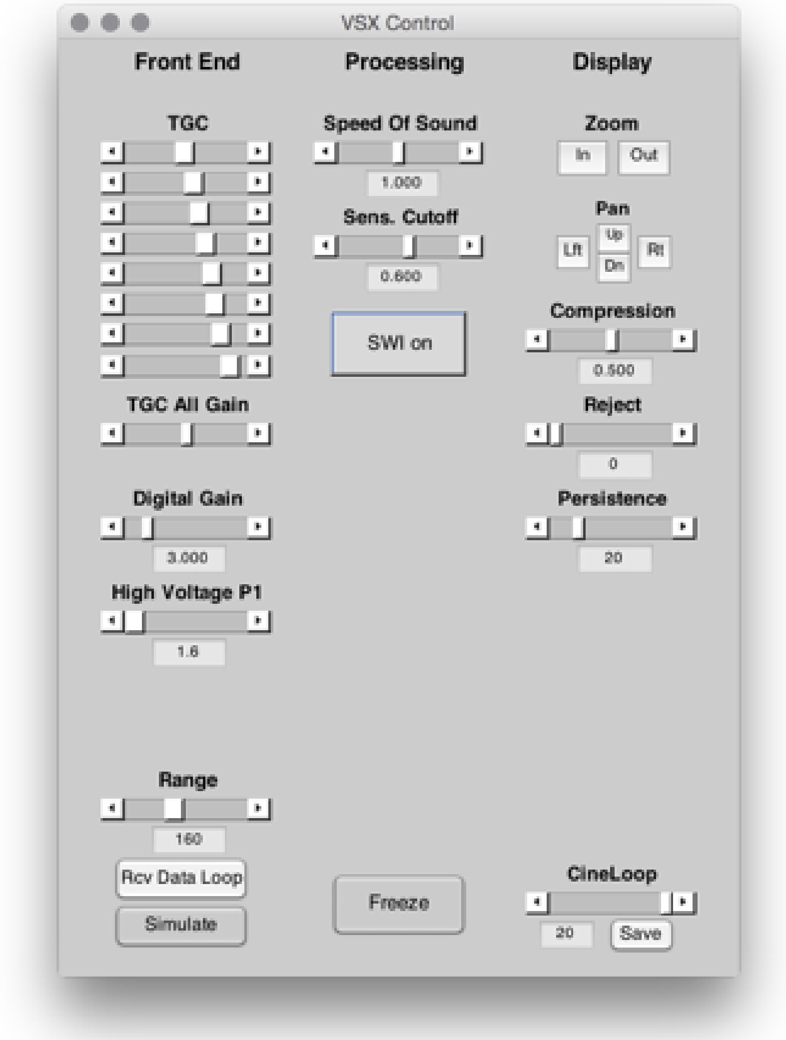
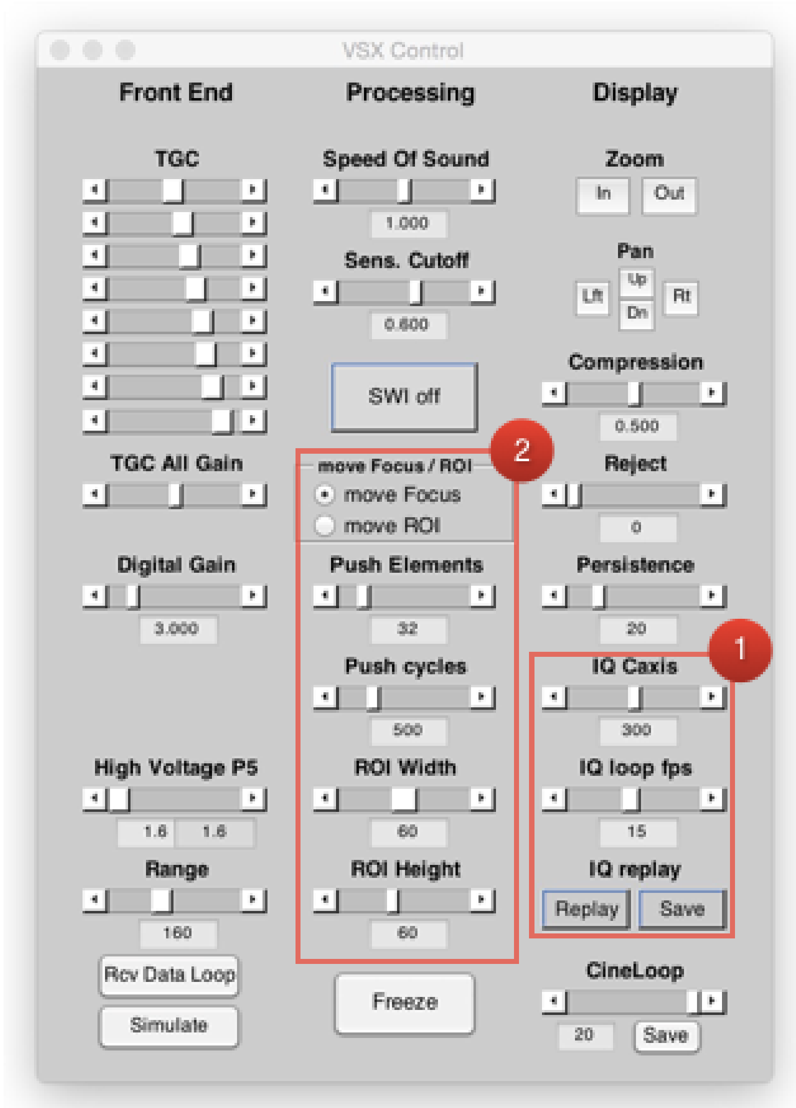


Click “SWI on” to turn on shear wave visualization mode. Since profile 5 will be used for imaging as well, only High Voltage P5 will be shown in the GUI.

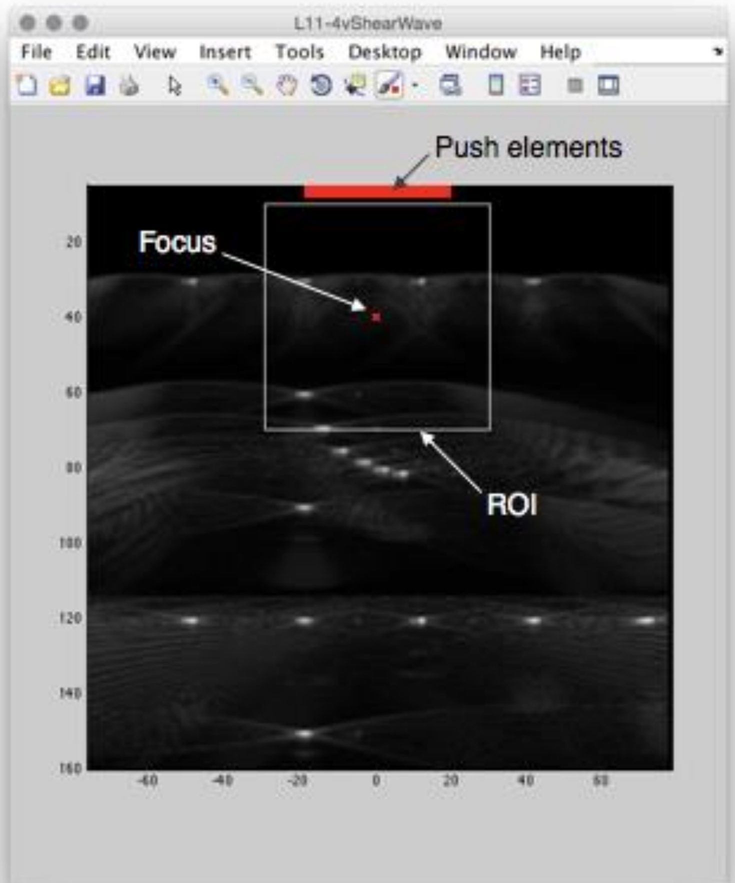


Some uicontrols and a new figure entitled “ShearWaveVisulization” will appear. The uicontrols are classified into two parts:

1. Shearwave (IQ) movie adjustment, including frame rate, Caxis for brightness adjustment, move replay, and save (as .avi) file.
2. Push / ROI adjustment, including push elements, push cycles, move focus and region of interest (ROI) adjustment.

In addition, the focus, elements for pushing, and ROI will be shown on original Bmode image. As shown below, the red rectangle indicates the location of all elements used for pushing. The focus was shown as a red cross and the white ROI represents the SWI region which will be displayed in the “ShearWaveVisulization” figure.

The default option in the "move Focus/ROI" group is 'move Focus', so the user is able to set push focus at a desired location by clicking left mouse button on the figure. The transmit current limit at the new focus will be calculated by TXLimitCheck function first. If the results failed to pass the limit, the focus will be moved back to its old location. If the results pass the limit, the red cross will move to a new focus location. Usually, locating the focus close to the center of the transducer is recommended.



If the user selects "move ROI", the focus will not be adjustable. Instead of changing the focus, the user can drag the ROI to another position to visualize wave propagation. As shown below, the ROI edge becomes dashed line when a user drags the ROI. New ROI will be determined after mouse button release.

If a user increases the push elements, the corresponding change will be displayed in the red rectangle after passing the TXLimitCheck, but the adjustment of push cycles will not be displayed. ROI Width and ROI Height are used to adjust the ROI size.

Instead of only one InterBuffer frame shown in most example scripts, the IQ data of each frame will be stored in this script. Therefore, the Recon and ReconInfo need to be defined correctly.

Usually the shear wave propagation is too fast to be visualized, so the “replay” button is used to replay the the wave propagation. By clicking replay after freeze, the user is able to adjust the frame rate (IQ loop fps) and brightness scale (IQ Caxis) to optimize the movie of shear wave propagation. Finally, the movie file can be saved to a desired folder by clicking save button.

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Verasonics, Inc.
11335 NE 122nd Way, Suite 100
Kirkland, WA 98034
USA